

Vasanth Modepalli

Java Developer | 4 Years Experience
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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Java Developer with 4 years of experience designing, building, and optimizing backend systems and microservices for large-scale platforms. Proficient in Java, Spring Boot, REST APIs, Oracle SQL, and backend automation. Demonstrated expertise in API design, database query optimization, dynamic service integrations, and critical production issue resolution.

TECHNICAL SKILLS

Languages: Java 8/17, SQL, PL/SQL, Python, Kafka
Frontend: React, JavaScript (ES6+), HTML5, CSS3
Backend: Spring Boot, Spring MVC, REST APIs, Microservices, Hibernate, Spring Data JPA, FastAPI
Databases: Oracle, MySQL, PostgreSQL, Redis
Tools: Git, Postman, Maven, IntelliJ, Docker, Linux, Cron Jobs
Concepts: Backend Automation, RCA, API Integration, SQL Optimization, Data Reconciliation, Production Debugging

PROFESSIONAL EXPERIENCE

Infosys Ltd. Jan 2024 - Present
Senior Systems Engineer — Java Full Stack Developer

- Designed and developed backend modules in Java and Spring Boot for taxpayer-facing government portal services, contributing to features used by millions of users across high-traffic filing cycles.
- Built React-based frontend forms for taxpayer data entry workflows on the income tax portal, integrating UI components with backend REST APIs for structured, validated data submissions.
- Engineered REST API enhancements and backend fixes across service and database layers — analyzed API contracts, service interactions, and DB schemas to implement correct, production-safe changes independently.
- Architected and built an automated reconciliation pipeline using Python and Oracle SQL that generates JSON payloads and triggers downstream feed processing — reducing a multi-hour manual process to under 5 minutes for 4k–5k records.
- Designed and implemented 22 dynamic email notification templates using FreeMarker-based HTML, integrating them end-to-end with backend communication services and notification dispatch logic.
- Optimized complex SQL queries and stored procedures, improving backend data retrieval performance by approximately 25% during peak filing loads — identified bottlenecks through query plan analysis and index tuning.
- Diagnosed and resolved critical production issues including feed failures, API contract mismatches, and database inconsistencies — performed structured RCA by tracing service logs, SQL execution paths, and interface payloads.
- Collaborated on backend service design decisions, reviewing data flow across microservices and ensuring consistency between API responses and downstream system expectations.

Infosys Ltd. Nov 2022 - Jan 2024
Systems Engineer — Java Backend Developer

- Developed and maintained backend components for critical income tax portal workflows, ensuring system correctness during high-volume filing periods with concurrent user spikes.
- Designed and executed API contract validation across backend services, ensuring correct data flow and catching integration regressions before they reached production.
- Built SQL-based diagnostic utilities to trace data inconsistencies across backend services, enabling faster resolution of data integrity issues in production.
- Identified and resolved root causes of backend failures across Java services, SQL layers, and external interface payloads — reducing repeat incidents through permanent code and query-level fixes.

KEY PROJECT

Scalable URL Shortener Service Java 17, Spring Boot, MySQL, Redis, Docker

- Designed and built a production-grade RESTful URL shortener service using Base62 encoding with collision-safe ID generation.
- Implemented a Redis caching layer with TTL-based expiry to serve high-frequency redirects with sub-millisecond latency, reducing database load significantly.
- Built an analytics module to track click counts, referrer patterns, and usage trends per shortened URL.
- Enforced input integrity using DTO-level validation and designed a centralized exception handling layer with structured error responses.
- Containerized the full application stack using Docker for environment-consistent local and deployment setup.

EDUCATION

B.Tech Mechanical Engineering 2018–2022
Sasi Institute of Technology and Engineering

CERTIFICATIONS

- Infosys Certified Spring Boot Developer
- Infosys Certified Java SE8 Developer

- Solved 150+ DSA problems on LeetCode